Wildlife Species Classification Using Deep Learning

Leveraging computer vision and CNN architectures to identify wildlife species from camera trap imagery with advanced active learning techniques.

Project Overview

Core Objective

Develop a robust CNN-based model to automatically classify wildlife species from camera trap images, addressing real-world challenges like occlusion, noise, and class imbalance.

Key Innovation

Integrate active learning to iteratively improve model performance by identifying uncertain predictions and strategically selecting samples for manual annotation.

The Caltech Camera Traps Dataset

243K

Total Images

Captured from 140 camera locations across diverse habitats

106K

Training Set

Images from 65 different camera locations

12.7K

Validation Set

From 10 new locations unseen during training

124K

Test Set

Images from 65 unique camera locations

Dataset includes COCO-style annotations with camera-trap specific metadata: location, datetime, sequence ID, and frame numbers. Over 57,000 images include precise bounding box annotations.

Dataset Capabilities

Binary Classification

Animal present vs. empty frame detection for efficient filtering

Species Classification

Class-level labels across multiple species categories

Note: The final analysis is based on, a subset of CCT, i.e., CCT - 20.

Data Preprocessing & Augmentation Strategy

Spatial Transformations

Random crop, horizontal flip, and rotation (±15°) to simulate varied camera angles and animal positions.

Color Adjustments

Brightness, contrast, and saturation jittering to handle varying lighting conditions across different times of day.

Noise & Blur

Blur to simulate real-world camera imperfections and motion blur.

Occlusion Handling

Cutout and random erase methods to train robustness against partially hidden animals and vegetation obstruction.

Training Methodology & Optimization

01

Progressive Fine-Tuning

Begin with frozen backbone layers, fine-tune classification head, then progressively unlock deeper layers for full network optimization.

02

Class Imbalance Mitigation

Implement weighted sampling and focal loss to address rare species, ensuring balanced learning across all categories.

03

Adaptive Learning Rate

Cosine annealing and ReduceLROnPlateau schedulers dynamically adjust learning rate based on validation performance.

04

Regularization Techniques

Early stopping, label smoothing, and ensemble predictions prevent overfitting and improve generalization to unseen camera locations.

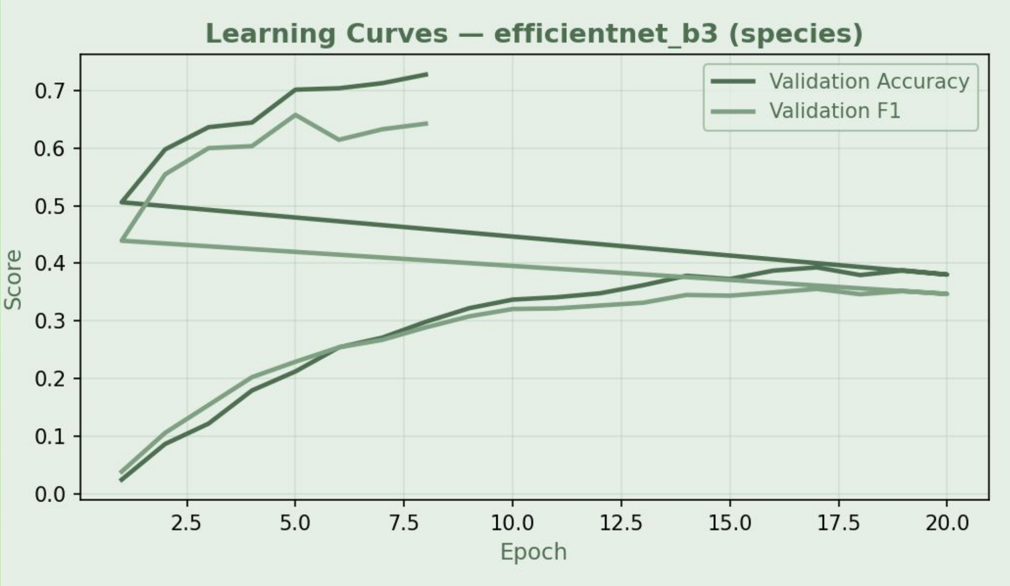
Training Dynamics: Species Classification

- **Stable learning behavior**
 Validation accuracy and F1 steadily improve during early epochs, showing the model is learning meaningful species-specific features.

- **No overfitting detected**
 The gap between Validation Accuracy and Validation F1 remains consistent, with no sudden divergence, indicating balanced generalization.

- **Fine-tuning improves performance**
 After the initial frozen-backbone phase, fully unfreezing EfficientNet-B3 leads to smoother and more consistent improvements across both metrics.

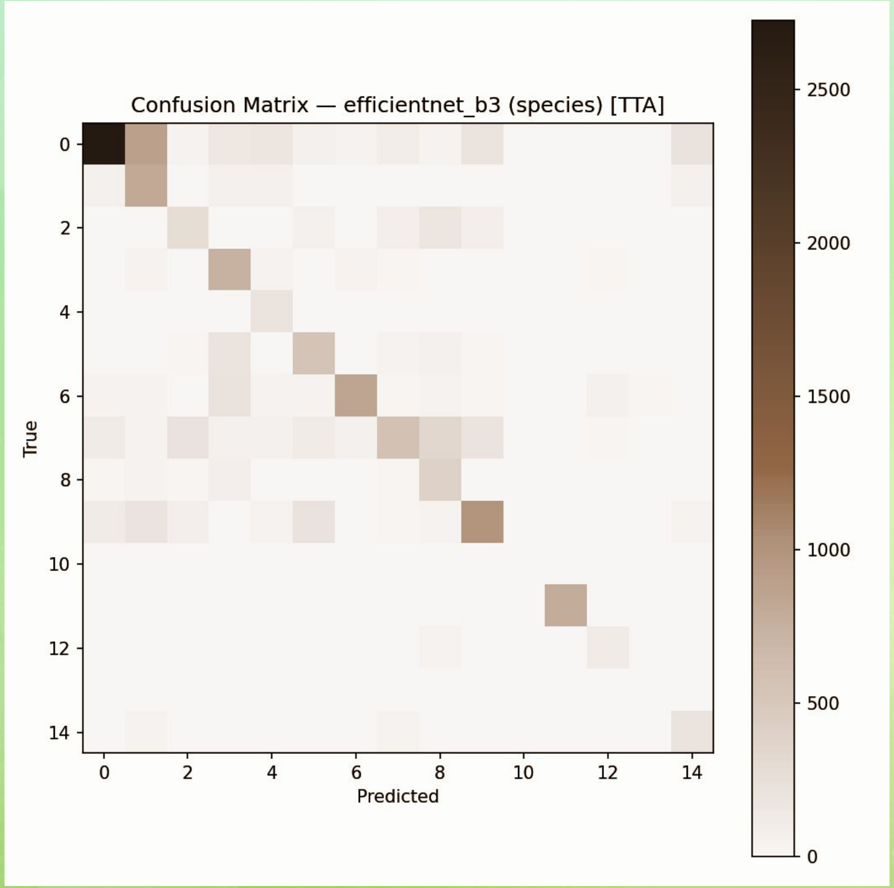
- **F1 grows more slowly than accuracy**
 Due to class imbalance and rare species (e.g., badger, skunk), F1 increases at a slower rate than accuracy which is an expected behavior.



Species Confusion Matrix

Strong diagonal (high correct predictions)
Most species fall cleanly along the diagonal, indicating the model reliably identifies the majority of classes.

- **Typical confusion patterns**
  A few clusters of misclassification appear where species share similar size, silhouette, or fur patterns.
  Most notable groups include:
  - **Rabbit ⇔ Squirrel** (small mammals with similar posture)
  - **Cat ⇔ Dog** (domestic animals with overlapping shapes)
  - **Skunk ⇔ Small dark mammals** in low-light conditions
- **Class imbalance influences mistakes**
  Rare species (e.g., **badger**, **skunk**) produce lighter, more scattered rows due to low sample counts, leading to occasional random misclassifications.



Top Misclassifications

Confusions concentrate in visually similar species

These species share overlapping silhouettes in low-resolution camera-trap images.

Lighting conditions heavily influence mistakes

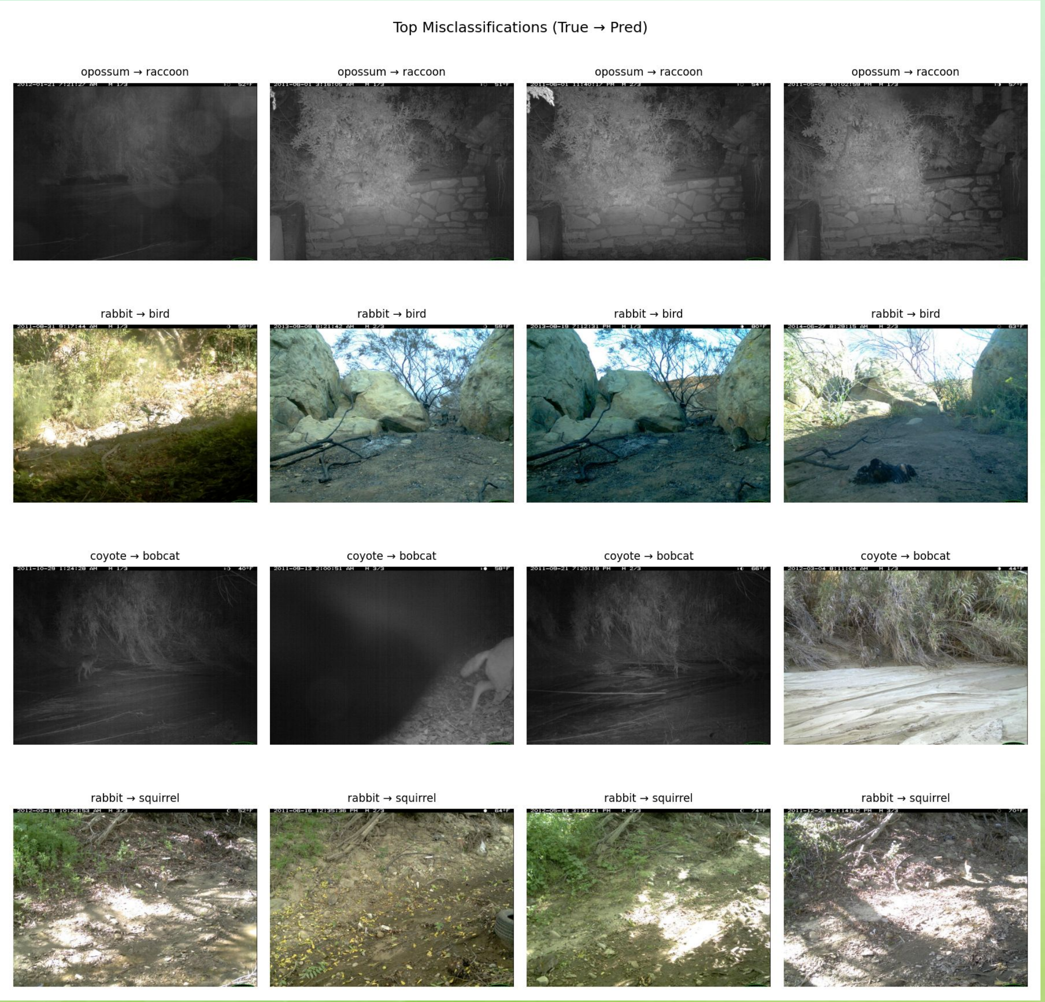
A large portion of misclassifications come from **nighttime infrared images**, where:

- Fur patterns disappear
- Contrast is extremely low
- Key features become indistinguishable

Environmental clutter & partial visibility contribute to errors

Misclassified images often include:

- Vegetation occluding the body
- Motion blur
- Small animals partially off-frame Heavy shadows or overexposure



Per-Class Performance Analysis

Class	Precision	Recall	F1-Score	Support
opossum	0.894133	0.603006	0.720264	4524
raccoon	0.395732	0.77937	0.524928	1047
squirrel	0.448105	0.412747	0.4297	659
bobcat	0.492608	0.817168	0.614675	897
skunk	0.317376	0.92268	0.472296	194
dog	0.532683	0.646154	0.583957	845
coyote	0.844444	0.630468	0.721934	1326
rabbit	0.663562	0.324232	0.435613	1758
bird	0.359195	0.65445	0.463822	573
cat	0.669649	0.596201	0.630794	1632
badger	0.6	0	0	4
car	0.989937	0.994943	0.992434	791

Key Insights

Strong overall performance with 3 classes >70% F1-scores (opossum, coyote, car)

"Car" classification near-perfect (99% across all metrics) - easiest to distinguish

Severe class imbalance affecting performance: badger (4 samples) vs. opossum (4,524 samples)

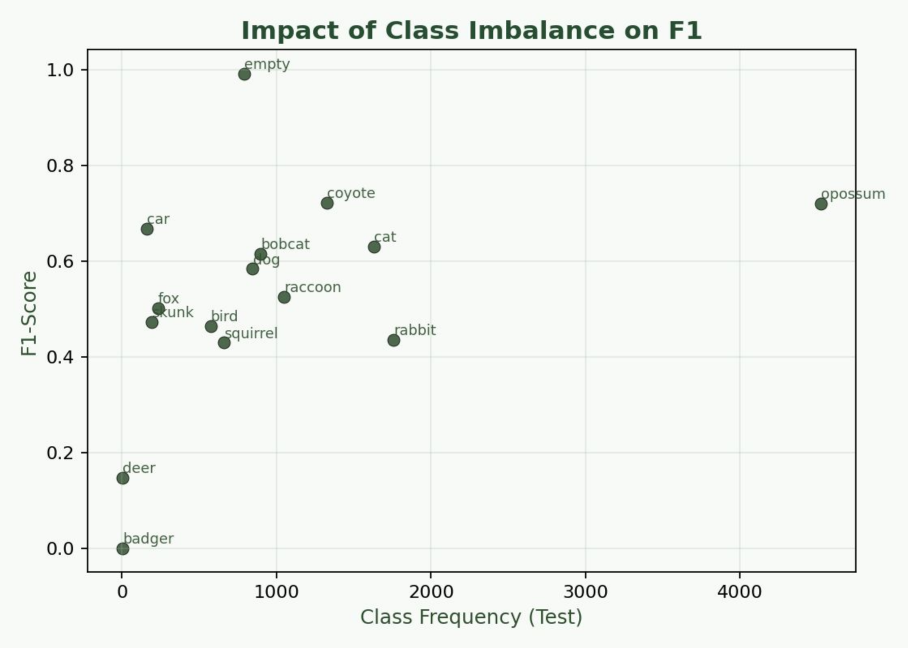
Key Challenges

Skunk: High recall (92%) but low precision (32%) → model overclassifies as skunk

Badger: Zero recall with minimal training data (n=4) → needs more samples

Rabbit: Low recall (32%) despite decent precision → frequently misclassified as other animals

Class Imbalance vs F1



Clear relationship: more samples → higher F1

- ➔ Classes with large test-set frequency (e.g., **opossum**, **coyote**, **car**) achieve the **highest F1-scores**.
- ➔ This shows the model learns well when enough training data exists.

Middle-frequency species cluster around moderate F1

- ➔ Species such as **cat, dog, raccoon, bobcat, fox** have moderate frequency and group around **0.5–0.65 F1**, reflecting partial but imperfect learning.

Class imbalance is a major performance bottleneck

- ➔ Even with class-weighted loss and augmentation, the model struggles to form strong decision boundaries for tail classes.

This directly motivates:

- better resampling
- targeted data collection
- synthetic augmentation
- future active learning

Grad-CAM Interpretability: Model Explanations via Grad-CAM

Model focuses on key anatomical regions

The Grad-CAM heatmaps show that EfficientNet-B3 attends strongly to the **core body regions** of each species:

- torso and wings for **bird**
- hind legs and tail for **coyote**
- shoulder and upper body region for **bobcat**

This indicates the network is using **semantically meaningful cues**, not random background patterns.

Class-specific features are highlighted correctly

- **Bird:** texture + color patches in the foreground vegetation
- **Coyote:** strong focus on muscular hind area and tail silhouette
- **Bobcat:** attention to patterned fur regions and posture

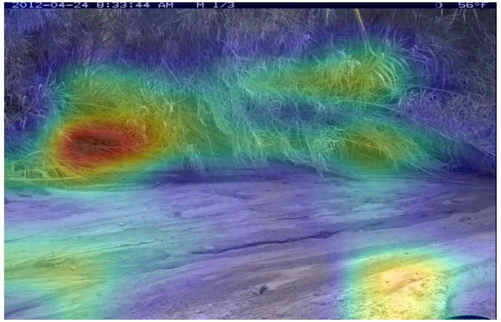
This aligns with how human observers would distinguish these animals.

Interpretability confirms model reliability

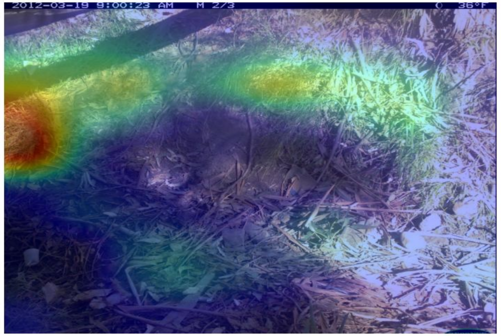
Grad-CAM results show the network is **not relying on spurious artifacts** such as timestamps, corners, or empty patches.
This increases trust in the learned representations and supports the validity of model predictions.

Grad-CAM — coyote

Grad-CAM — bobcat

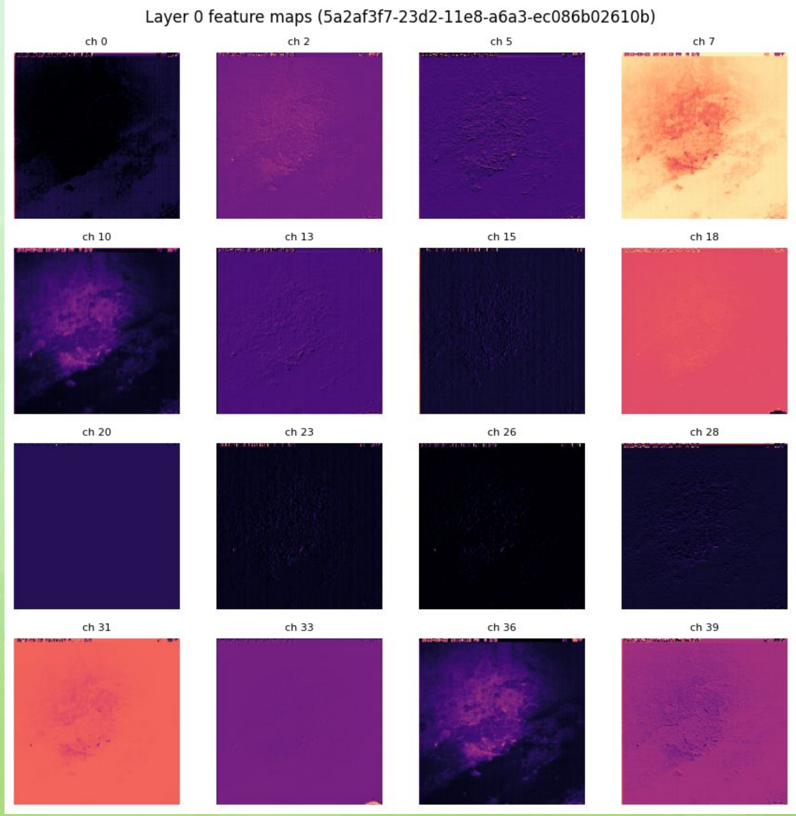


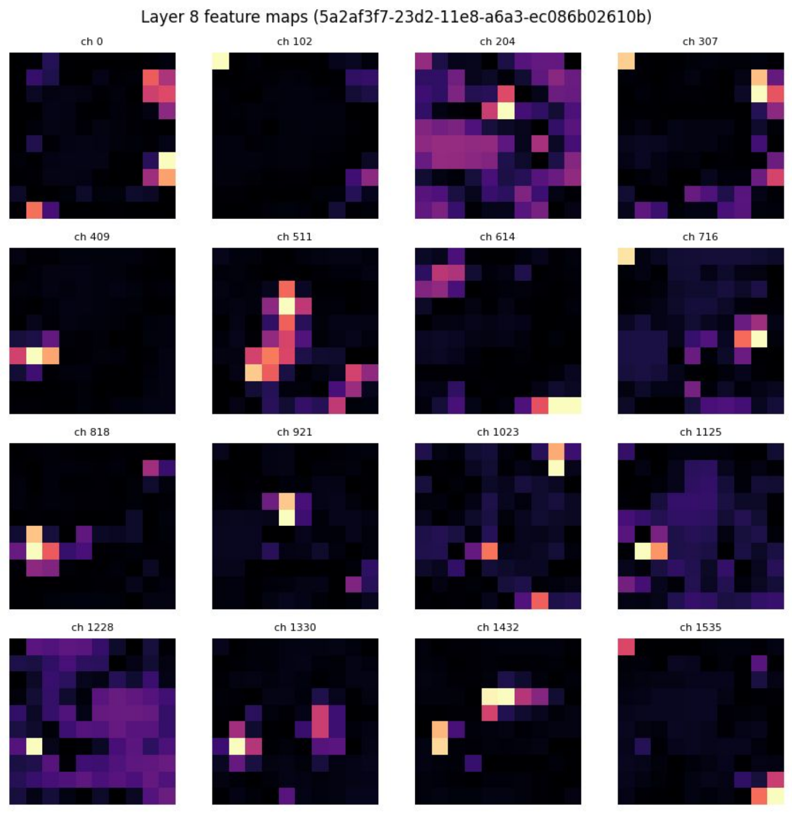
Grad-CAM — bird



Feature Maps (Early vs Late)

Hierarchical progression confirms correct learning





- ➔ Early layers learn simple textures and edges
- ➔ Later layers focus on meaningful, localized regions

Findings

n_samples: 14647
Accuracy: 0.6224
Macro-F1: 0.5273
Weighted-F1: 0.632
Precision_macro:0.5104
Precision_weighted: 0.7005

Best & Worst Classes

Best: opossum, coyote, car

Hardest : badger, rabbit, skunk

Limitations

01

Data size

The dataset size was very large and took a lot of compute for our computers. For this analysis we decided to use a subset of the data.

02

Class Imbalance

The dataset was imbalanced with common species like opossum dominating, while rare species had very few samples.

03

Unpredictability of the data

There is a lot of uncertainty and things that we cannot predict within nature. While augmentation and adjustments to the training set helps, there is no way to full simulate nature.

Questions?